Template Week 5 – Operating Systems

Student number: 591491

Assignment 5.1: Unix-like

a) Find out what the difference is between UNIX and unix-like operating systems?

Unix is echt een besturingssysteem tak dat gecertificeerd Unix is en unix-like lijkt op Unix maar heeft daar geen licentie voor en is ook niet goedgekeurd door Unix.

b) Study the image above named UNIX timeline. Find out who Ken Thompson, Dennis Ritchie, Bill Joy, Richard Stallman, and Linus Torvalds are and what they have contributed to the development of UNIX or unix-like systems and to IT in general. **TIP!** English-language sources often contain more detailed information about these individuals.

Ken Thompson en Dennis Ritchie hebben UNIX gemaakt toen ze bij Bell Labs werkten.

Ken Thompson ontwikkelde ook de B programmeertaal die veel invloed heeft gehad op C.

Dennis Ritchie was de hoofdmaker van de C-programmeertaal die was bedoeld voor de ontwikkeling van Unix.

Bill Joy ontwikkelde de BSD-versie van UNIX, die veel belangrijke tools en netwerkmogelijkheden bracht.

Billy Joy creëerde ook de C-shell (csh) en had invloed op de ontwikkeling van Java.

Richard Stallman startte het GNU-project om een gratis en open-source besturingssysteem te creëren als alternatief voor Unix.

Linus Torvalds schreef de Linux kernel die we kennen en werkt daar nog steeds aan

Ook heeft Linus Torvalds GIT geschreven in 2005.

c) What is the philosophy of the GNU movement?

Om Software pakketten open source te maken zodat iedereen het kan lezen of aanpassen, in tegenstelling tot Unix die heel veel closed source houdt.

d) Does Ubuntu as a Linux operating system conform to the philosophy of the GNU movement? Please explain your answer.

Ja, behoorlijk erg wel. Er staat veel GNU software op en kan veel op gedownload worden. Ubuntu is ook open source en iedereen mag er aanpassingen aan maken. Wel is er ubuntu pro subscription voor extra lange beveilingssupport, maar dit is ook gebouwd op open source tools en is redelijk conform aan de GNU movement volgens mij, ookal kan het ook zijn dat kleine gedeeltes hiervan dat niet zijn.

e) Find out what is the Windows Subsystem for Linux?

WSL is een kleine Linux kernel die Windows tegelijkertijd kan draaien. Dit maakt het mogelijk om Linux bij naast je Windows tegelijk te draaien. Dit wordt veel gebruikt voor virtuele Linux machines op windows en containers te draaien.

f) Find out, which operating system family belongs to Android, iOS and ChromeOS?

ChromeOS is gebaseerd op Linux en hoort bij die familie

Android is ook Linux-based

En iOS is Unix-like

Assignment 5.2: Supercomputers and gameconsoles

a) Research on this site what supercomputers are used for and write a short summary of it: https://www.computerhistory.org/timeline/search/?q=Supercomputer

Supercomputers worden gebruikt voor ingewikkelde berekeningen en wetenschappelijk onderzoek. Ze doen taken die gewone computers niet kunnen, zoals het voorspellen van gegevens op hele grote datasets, het maken van datasets, het ophalen van gegevens uit hele grote datasets en het beveiligen van gegevens.

b) IBM is a company that has already built a number of supercomputers. One of them is IBM's Roadrunner. The CPU developed for this supercomputer was further developed at a later stage as the CPU for the PlayStation 3 console. Find out what a **PlayStation 3 cluster** is and what it was used for?

PS3 clusters zijn een soort super computers die zijn gebouwd van eerste generaties ps3. De cell processor van de ps3 was heel goed in parallelle berekeningen en had een lage prijs waardoor het redelijk rendabel was om veel ps3 te kopen, linux erop te zetten en er daarna een cluster van te bouwen.

c) You can build a supercomputer by putting a few computers together in a cluster. Here's what Oracle did with a collection of Raspberry Pi's, for example: https://blogs.oracle.com/developers/post/building-the-worlds-largest-raspberry-pi-cluster
What specific operating system is running on this cluster?

Oracle Linux for ARM in plaats van de normale raspbian

d) Does Oracle's Raspberry Pi supercomputer appear in the list of the 500 fastest supercomputers in the world? Make a logical decision for this, without going through the entire list. https://www.top500.org/lists/top500/list/2023/06/

Nee, hij staat er niet in. Ik ben naar de 500ste plek gegaan en zag dat er 40 000 cores in zaten met een totaal van 2 PFlops. Een raspberry pi 3 4 en 5 hebben maar 4 cores dus dat komt uit op 1050x4=4200 cores totaal. Ik verwacht ook niet dat die 2 PFlops hebben in totaal.

e) What CPU architecture is used for the PlayStation 5 and Xbox Series X?

8-core x86-64 AMD Zen 2 CPU architecture

What operating systems run on these consoles?

Ps5: een custom Unix-like besturingsysteem met een kernel gebaseerd op FreeBSD

xbox: een custom os gebaseerd op de Windows 11 kernel

What conclusion can you draw from the answer to the previous question?
Ps5 is unix-like en xbox Windows. Ook betekent het dat de performance afhangt van de os en de optimalisatie van de game, maar het zou dus hetzelfde kunnen zijn vanwege bijna dezelfde hardware.

Assignment 5.3: Working with Windows

Take relevant screenshots of the assignments below

a) Practice for about 10 minutes with the ⊞ keyboard shortcuts combinations, skip the general shortcuts in this exercise. Take a look at which screens are opened.

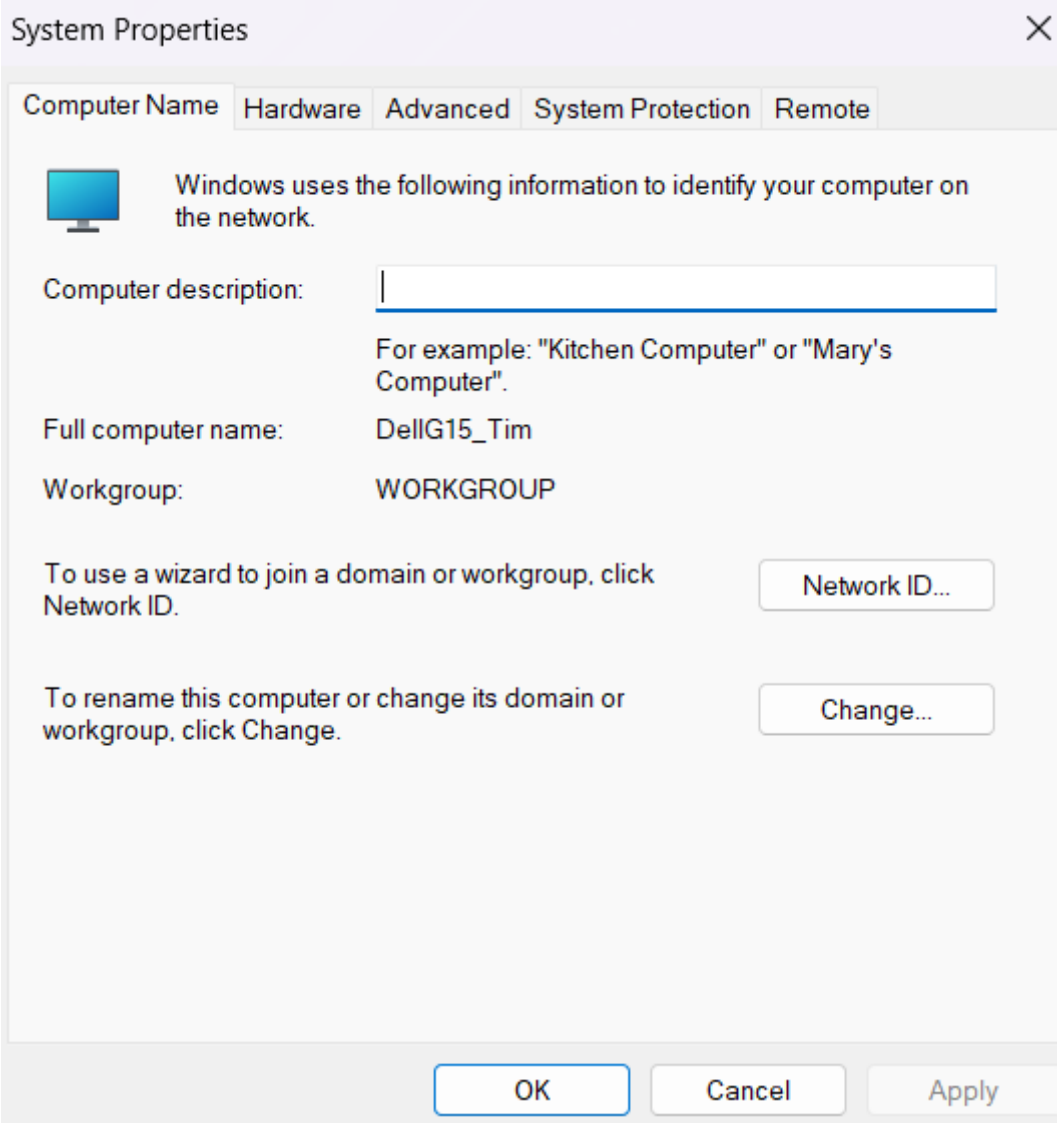
b) The file explorer can be opened with ⊞ + E, Which key combination could you also use?

Windows key + 1

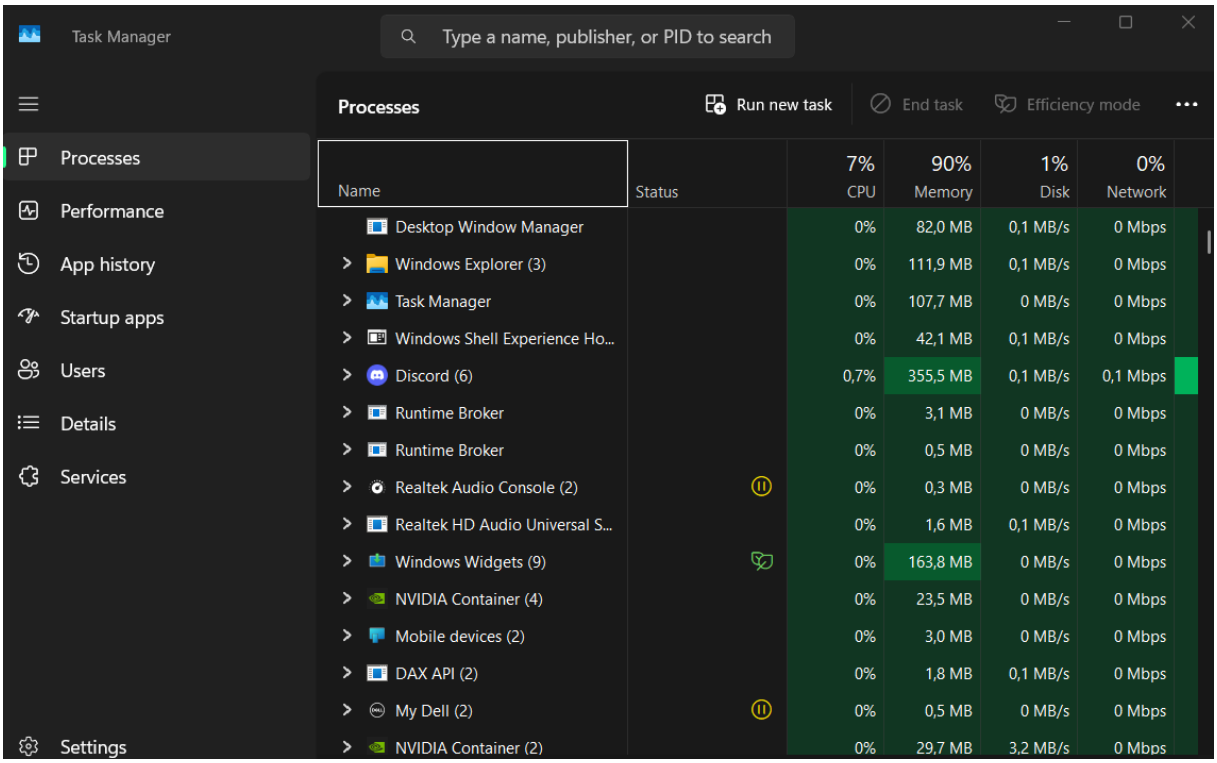
Of Windows key + R en daarna typ je explorer en druk je op enter

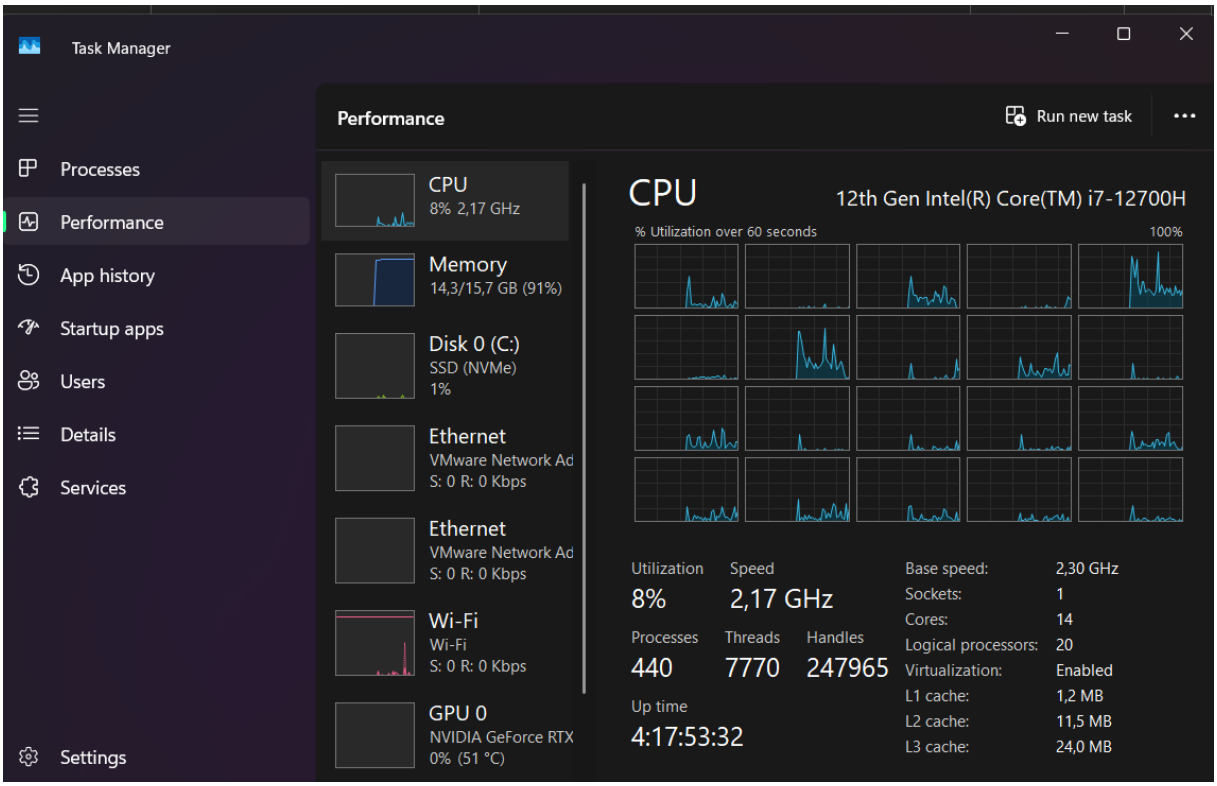
c) Open the system properties with a ⊞ key combination, take a screenshot of the open screen. Paste this screenshot into this template.

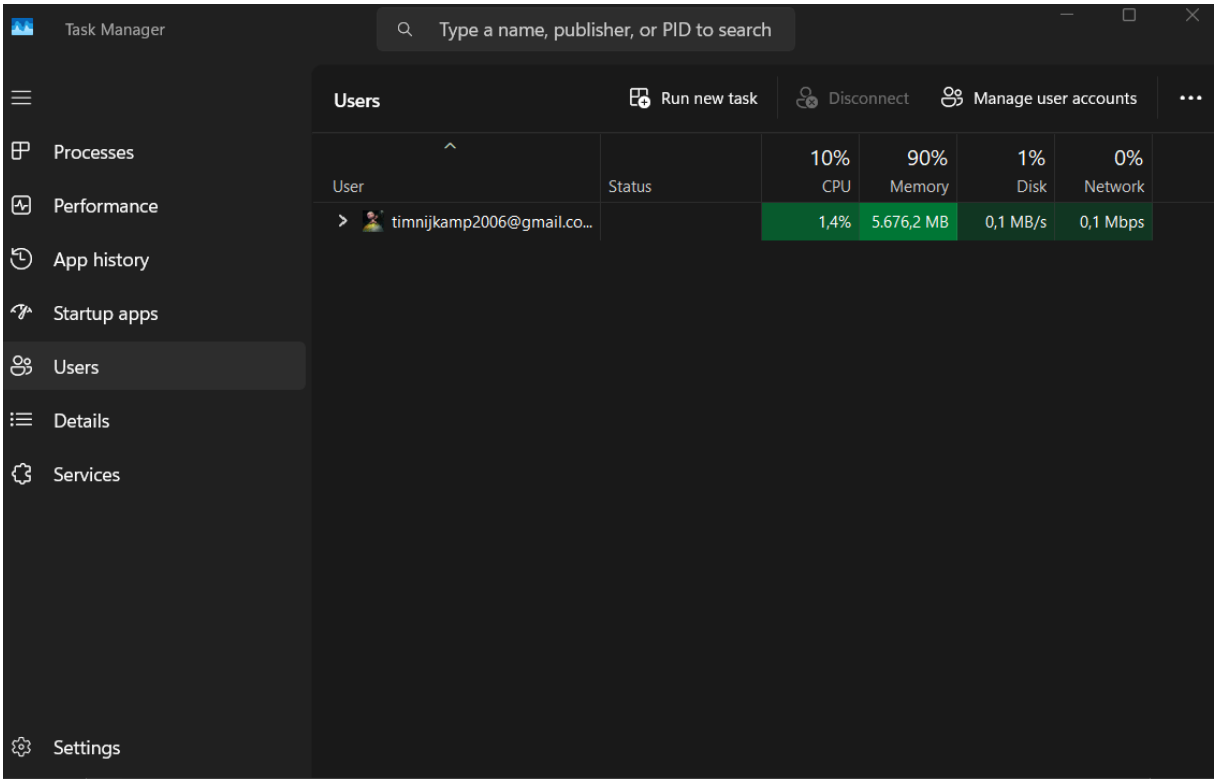
Ik heb geen pause key dus heb ik het handmatig geopend



d) Open task manager with a key combination. Take screenshots of the tabs: processes (shows active processes), performance, and users. Place these three screenshots in this template.







e) If you're giving a PowerPoint presentation and you connect your laptop to a projector, Windows can use the projector as a second screen. For example, you may have Outlook open on your first screen that you don't show over the projector, while the PowerPoint presentation is displayed on the projector, or the second screen. Which key combination should you use for this?

Win + P

f) If you leave the classroom for a while and you leave your laptop behind, it is wise to lock the screen. Your Apps will continue to run in the background. So, for example, if you're waiting for a download that takes a while, lock the screen and get a cup of coffee. Which key combination do you use for this?

Win + L

g) Open the Run screen with a key combination. On this screen, type CMD and press <enter>. Take a screenshot of this result and paste it into this template.

```
C:\WINDOWS\system32\CMD
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\timni>
```

Working in the File Explorer

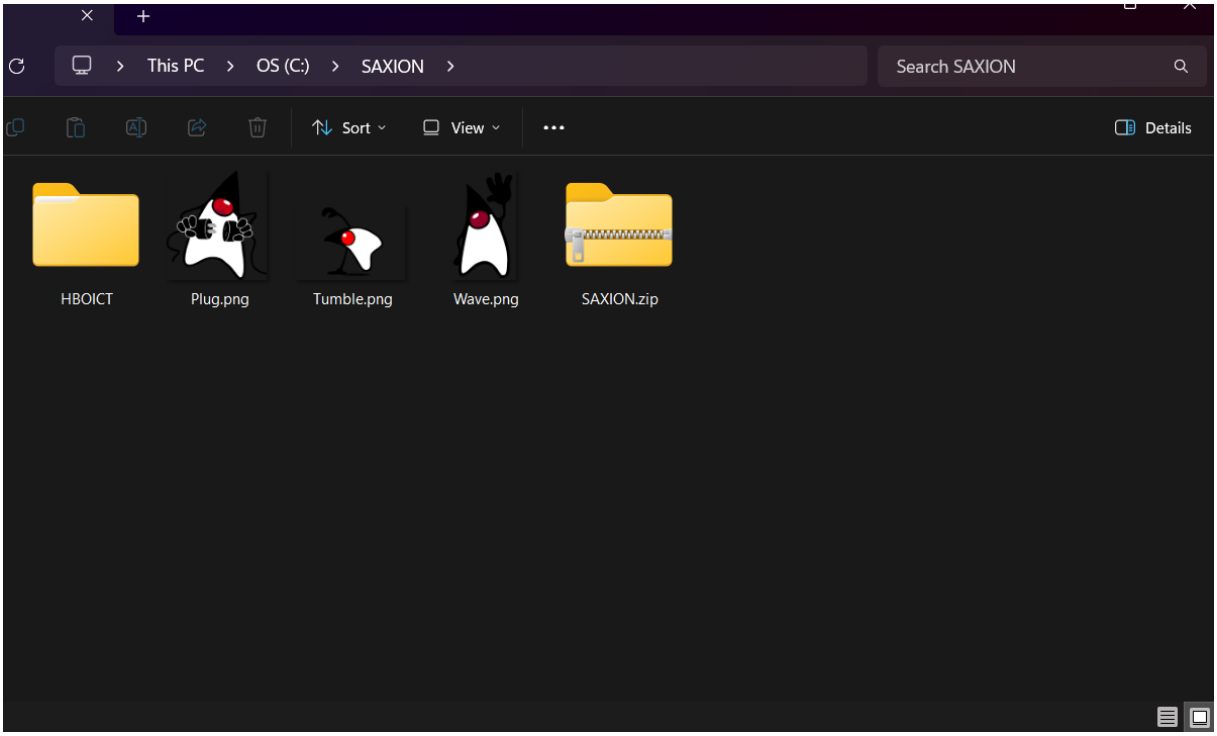
Relevant screenshots **copy** command:

```
C:\Windows\System32\cmd.e

C:\SAXION>copy C:\SAXION\Wave.png "C:\SAXION\HBOICT\YEAR1\QUARTILE1\Introductie Programmeren\"
        1 file(s) copied.

C:\SAXION>copy .\Wave.png ".\HBOICT\YEAR1\QUARTILE1\Introductie Infra\"
        1 file(s) copied.

C:\SAXION>copy .\Tumble.png "C:\SAXION\HBOICT\YEAR1\QUARTILE1\Organisatie en IT\"
        1 file(s) copied.

C:\SAXION>del ".\HBOICT\YEAR1\QUARTILE1\Introductie Infra\Wave.png"

C:\SAXION>copy .\Plug.png ".\HBOICT\YEAR1\QUARTILE1\Introductie Infra\"
        1 file(s) copied.

C:\SAXION>
```

Relevant screenshots **tree** command:

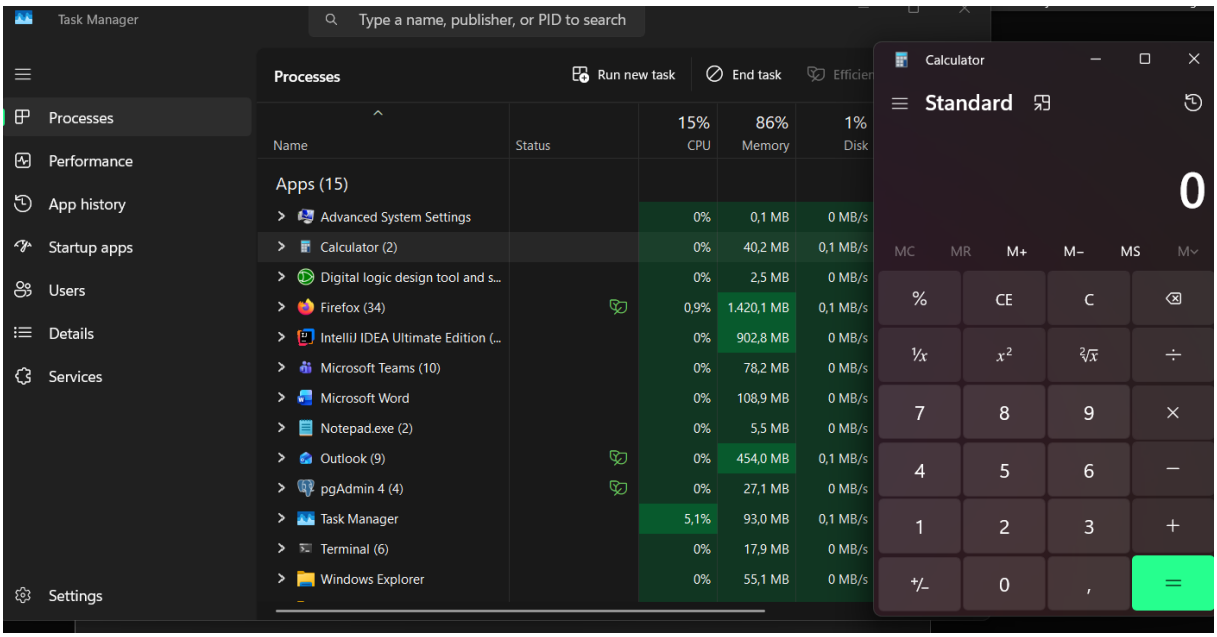
```
C:\SAXION>tree
Folder PATH listing for volume OS
Volume serial number is 9C54-66A7
C:.
└───HBOICT
    ├───YEAR1
    │   ├───QUARTILE1
    │   │   ├───Introductie Infra
    │   │   ├───Introductie Programmeren
    │   │   └───Organisatie en IT
    │   ├───QUARTILE2
    │   ├───QUARTILE3
    │   └───QUARTILE4
    ├───YEAR2
    │   ├───QUARTILE1
    │   ├───QUARTILE2
    │   ├───QUARTILE3
    │   └───QUARTILE4
    ├───YEAR3
    └───YEAR4

C:\SAXION>echo %username%
timni
```

Relevant screenshots in the file explorer of the folder c:\Saxion + created zip file.



Terminating Processes

Relevant Screenshots Task Manager Window:



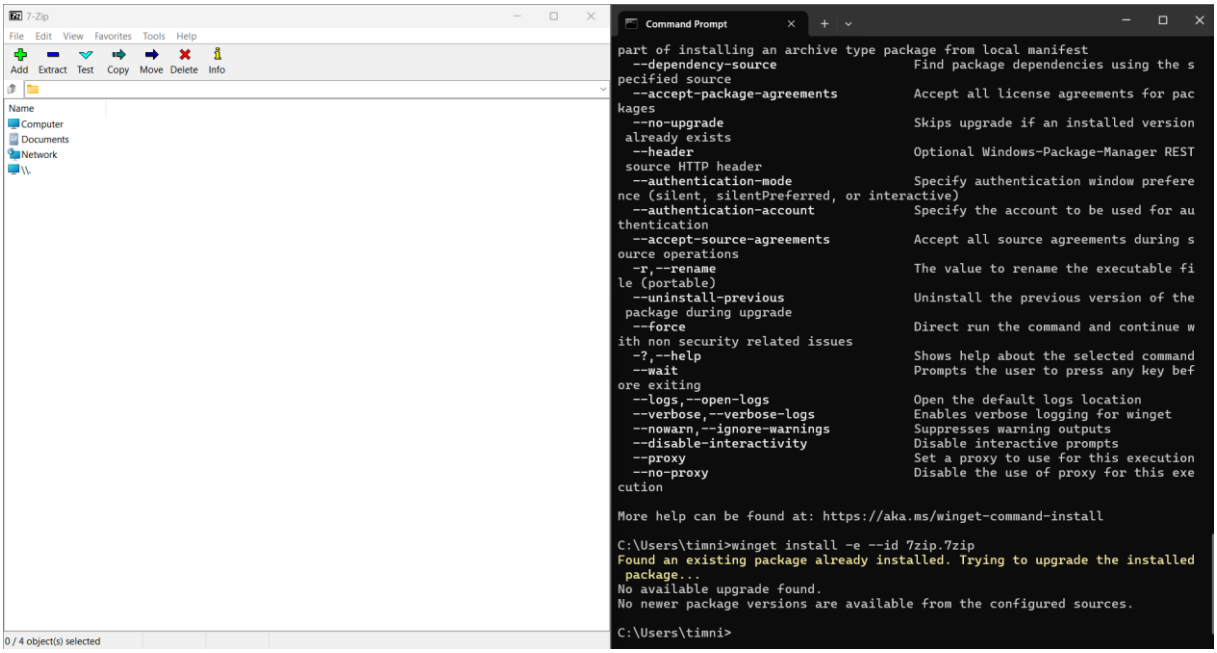
Install Software

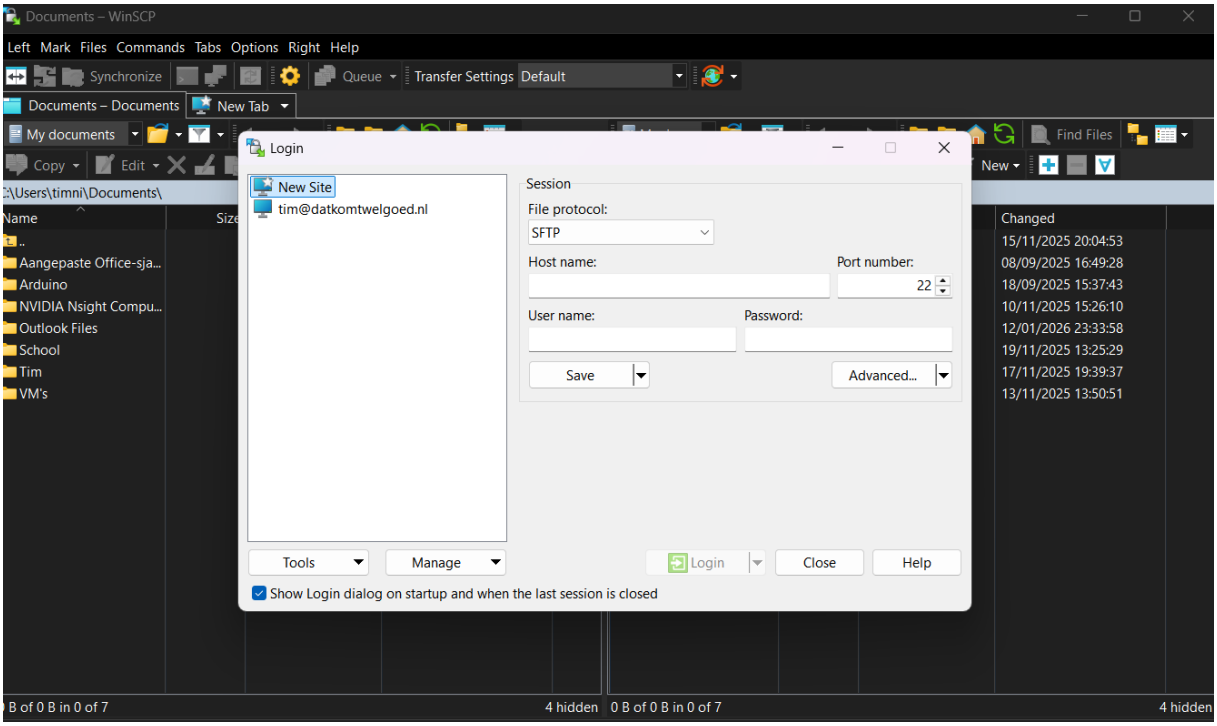
Relevant screenshots that the following software is installed with winget:

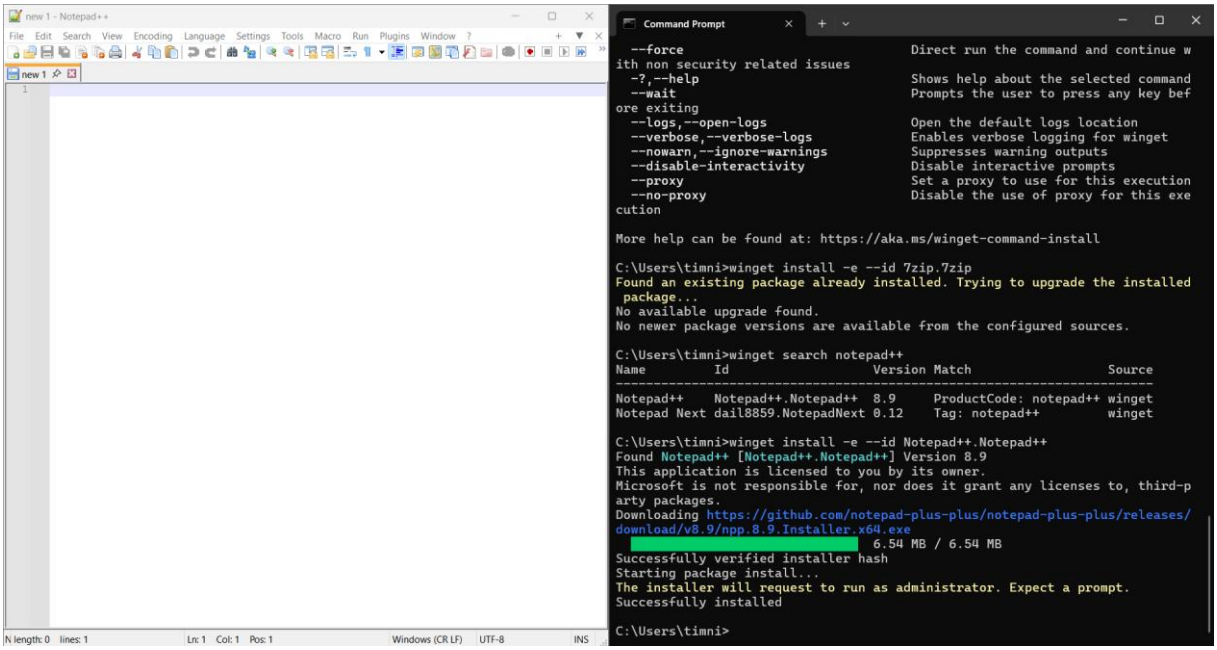
- WinSCP
- Notepad++
- 7zip

B winget install -e -id

Winget is het hoofd command met als optie install om iets te installeren. -e betekent dat hij het specifieke pakket aangeduid met -id alleen installeerd.




Documents – WinSCP
Left Mark Files Commands Tabs Options Right Help
Synchronize
Queue
Transfer Settings Default
Documents – Documents
New Tab
My documents
Find Files
Copy
Edit
New
C:\Users\timni\Documents\
Name
Size
Changed
..
Aangepaste Office-sja...
Arduino
NVIDIA Nsight Compu...
Outlook Files
School
Tim
VM's
15/11/2025 20:04:53
08/09/2025 16:49:28
18/09/2025 15:37:43
10/11/2025 15:26:10
12/01/2026 23:33:58
19/11/2025 13:25:29
17/11/2025 19:39:37
13/11/2025 13:50:51
Login
New Site
tim@datkomtwelgoed.nl
Session
File protocol:
SFTP
Host name:
Port number:
22
User name:
Password:
Save
Advanced...
Tools
Manage
Login
Close
Help
Show Login dialog on startup and when the last session is closed
B of 0 B in 0 of 7
4 hidden
0 B of 0 B in 0 of 7
4 hidden

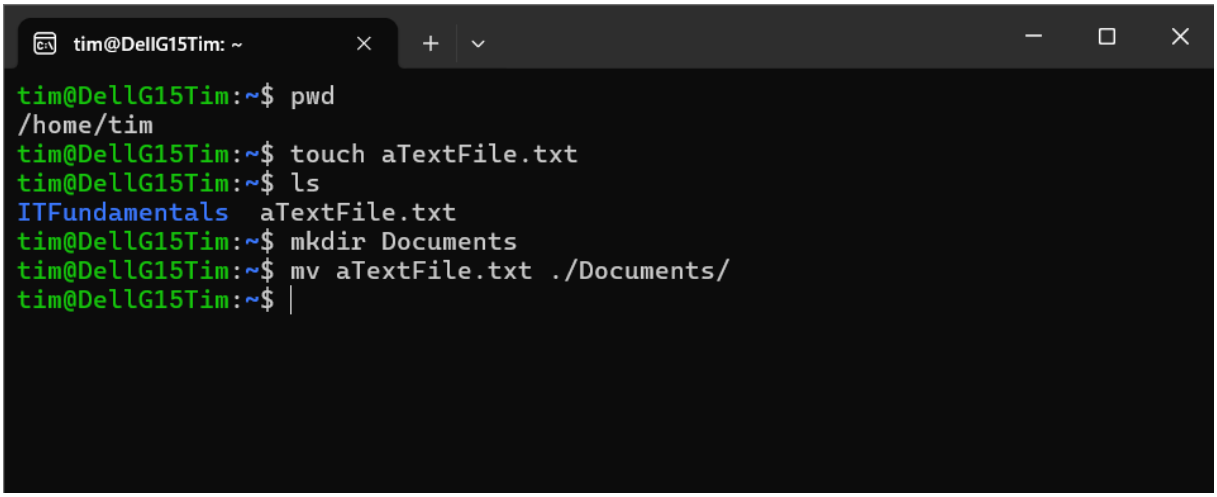

new 1 - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
new 1
N length: 0 lines: 1
Ln: 1 Col: 1 Pos: 1
Windows (CR LF) UTF-8 INS
Command Prompt
--force Direct run the command and continue w
ith non security related issues
-?,--help Shows help about the selected command
--wait Prompts the user to press any key bef
ore exiting
--logs,--open-logs Open the default logs location
--verbose,--verbose-logs Enables verbose logging for winget
--nowarn,--ignore-warnings Suppresses warning outputs
--disable-interactivity Disable interactive prompts
--proxy Set a proxy to use for this execution
--no-proxy Disable the use of proxy for this exe
cution
More help can be found at: https://aka.ms/winget-command-install
C:\Users\timni>winget install -e --id 7zip.7zip
Found an existing package already installed. Trying to upgrade the installed
package...
No available upgrade found.
No newer package versions are available from the configured sources.
C:\Users\timni>winget search notepad++
Name Id Version Match Source
Notepad++ Notepad++.Notepad++ 8.9 ProductCode: notepad++ winget
Notepad Next dail8859.NotepadNext 0.12 Tag: notepad++ winget
C:\Users\timni>winget install -e --id Notepad++.Notepad++
Found Notepad++ [Notepad++.Notepad++] Version 8.9
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-p
arty packages.
Downloading https://github.com/notepad-plus-plus/notepad-plus-plus/releases/
download/v8.9/npp.8.9.Installer.x64.exe
6.54 MB / 6.54 MB
Successfully verified installer hash
Starting package install...
The installer will request to run as administrator. Expect a prompt.
Successfully installed
C:\Users\timni>

Assignment 5.4: Working with Linux

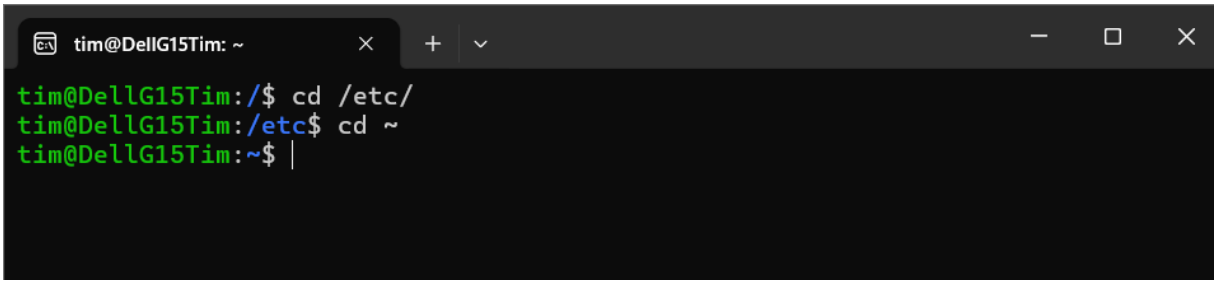
Relevant screenshots + motivation

copy

Ik heb geen grafische interface dus ik kan het helaas niet laten zien in de file explorer



Navigating



Het grootste verschil in bestandsstructuur is dat linux / heeft waar echt alles in staat. Je andere harde schrijven die staan als devices in /dev/ die je eerst moet mounten voordat je er makkelijk bij kan.

In /etc/ staan bijna alleen maar configuratie bestanden. Ik onthoud het als Everything Thats Configurable

Compress

Zie afbeelding hieronder

```
tim@DellG15Tim: ~/Documen
tim@DellG15Tim:~/Documents$ pwd
/home/tim/Documents
tim@DellG15Tim:~/Documents$ ls
aTextFile.txt
tim@DellG15Tim:~/Documents$ tar -cvf aTextFile.tar ./aTextFile.txt
./aTextFile.txt
tim@DellG15Tim:~/Documents$ ls
aTextFile.tar  aTextFile.txt
tim@DellG15Tim:~/Documents$ rm aTextFile.txt
tim@DellG15Tim:~/Documents$ ls
aTextFile.tar
tim@DellG15Tim:~/Documents$ tar -xvf aTextFile.tar
./aTextFile.txt
tim@DellG15Tim:~/Documents$ ls
aTextFile.tar  aTextFile.txt
tim@DellG15Tim:~/Documents$ gunzip
gzip: compressed data not read from a terminal. Use -f to force decompressio
n.
For help, type: gzip -h
tim@DellG15Tim:~/Documents$ gzip aTextFile.tar
tim@DellG15Tim:~/Documents$ ls -la
total 16
drwxr-xr-x 2 tim tim 4096 Jan 13 00:09 .
drwxr-x--- 6 tim tim 4096 Jan 12 23:56 ..
-rw-r--r-- 1 tim tim  157 Jan 13 00:05 aTextFile.tar.gz
-rw-r--r-- 1 tim tim   16 Jan 13 00:04 aTextFile.txt
tim@DellG15Tim:~/Documents$
```

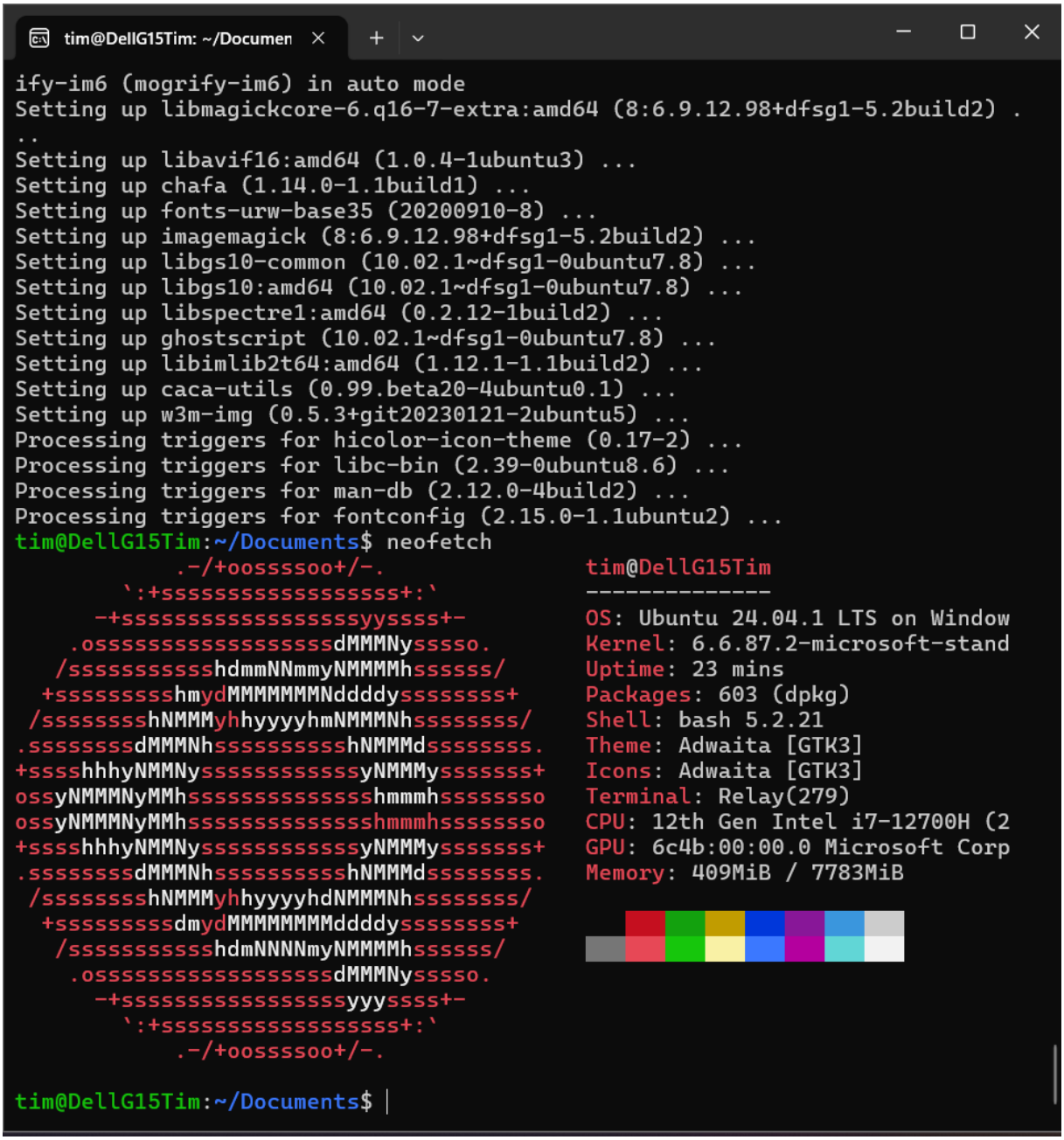
View processes

```
tim@DellG15Tim: ~/Documen

  0[0.0]   3[0.0]   6[0.0]   9[0.0]  10[0.0%] 13[0.0%] 16[0.0%] 19[0.0%]
  1[0.0]   4[0.0]   7[0.0]           11[0.0%] 14[0.0%] 17[0.0%]
  2[0.0]   5[0.0]   8[0.0]           12[0.0%] 15[0.0%] 18[0.0%]
Mem[||||||                 399M/7.60G] Tasks: 23, 12 thr, 0 kthr; 1 running
Swp[                         0K/2.00G] Load average: 0.00 0.00 0.00
                                       Uptime: 00:17:37

  Main   I/O
    PID USER       PRI  NI  VIRT   RES   SHR S  CPU%▽MEM%   TIME+  Command
      1 root        20   0 21592 11980  9100 S   0.0  0.2  0:00.44 /sbin/ini
      2 root        20   0  3072  1760  1760 S   0.0  0.0  0:00.00 /init
      6 root        20   0  3072  1760  1760 S   0.0  0.0  0:00.00 plan9 --c
      7 root        20   0  3072  1760  1760 S   0.0  0.0  0:00.00 plan9 --c
      8 root        20   0  3072  1760  1760 S   0.0  0.0  0:00.00 /init
     34 root        19  -1 50420 15040 14080 S   0.0  0.2  0:00.09 /usr/lib/
     82 root        20   0 25132  6240  4800 S   0.0  0.1  0:00.14 /usr/lib/
    164 systemd-re  20   0 21452 12640 10400 S   0.0  0.2  0:00.09 /usr/lib/
    168 systemd-ti  20   0 91020  7520  6720 S   0.0  0.1  0:00.07 /usr/lib/
    172 systemd-ti  20   0 91020  7520  6720 S   0.0  0.1  0:00.00 /usr/lib/
    174 root        20   0  4236  2400  2240 S   0.0  0.0  0:00.00 /usr/sbin
    175 messagebus  20   0  9588  4800  4320 S   0.0  0.1  0:00.04 @dbus-dae
    181 root        20   0 18012  8480  7520 S   0.0  0.1  0:00.06 /usr/lib/
    185 root        20   0  3160  1760  1760 S   0.0  0.0  0:00.00 /sbin/age
    201 syslog      20   0  217M  5280  4320 S   0.0  0.1  0:00.02 /usr/sbin
    204 root        20   0  3116  1760  1760 S   0.0  0.0  0:00.00 /sbin/age
    210 syslog      20   0  217M  5280  4320 S   0.0  0.1  0:00.00 /usr/sbin
    211 syslog      20   0  217M  5280  4320 S   0.0  0.1  0:00.00 /usr/sbin
    212 syslog      20   0  217M  5280  4320 S   0.0  0.1  0:00.00 /usr/sbin
    277 root        20   0  3076   800   800 S   0.0  0.0  0:00.00 /init
    278 root        20   0  3092   960   960 S   0.0  0.0  0:00.06 /init
    279 tim         20   0  6600  5760  3680 S   0.0  0.1  0:00.19 -bash
    280 root        20   0  6820  4320  3680 S   0.0  0.1  0:00.00 /bin/logi
    306 tim         20   0 20300 11040  9120 S   0.0  0.1  0:00.07 /usr/lib/
    307 tim         20   0 21144  3376  1760 S   0.0  0.0  0:00.00 (sd-pam)
    330 tim         20   0  6072  4960  3360 S   0.0  0.1  0:00.01 -bash
    659 root        20   0  361M 20160 17280 S   0.0  0.3  0:00.02 /usr/libe
    660 root        20   0  361M 20160 17280 S   0.0  0.3  0:00.00 /usr/libe
    661 root        20   0  361M 20160 17280 S   0.0  0.3  0:00.00 /usr/libe
    662 root        20   0  361M 20160 17280 S   0.0  0.3  0:00.00 /usr/libe
F1Help  F2Setup F3SearchF4FilterF5Tree  F6SortByF7Nice -F8Nice +F9Kill  F10Q
```

Bovenaan zie je de cores/threads. Zoals je hier kan zien zijn dit er 20. Ook kan je per thread zien hoeveel die in gebruik is. Daaronder zie het het swap geheugen en RAM. Ook zie je de avg load en de uptime.

In de lijst zie je precies per process hoeveel die verbruikt, welk process het is en door welke user het draait en nog wat andere dingen zoals het process identificatie nummer (PID)

Install software



Het laat wat info over mijn systeem zien

Assignment 5.5: Users and permissions on Linux

Relevant screenshots + motivation

```
tim@DellG15Tim: ~/hello
Setting up w3m-img (0.5.3+git20230121-2ubuntu5) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for fontconfig (2.15.0-1.1ubuntu2) ...
tim@DellG15Tim:~/Documents$ neofetch
            .-/+oossssoo+/-.               tim@DellG15Tim
        `:+ssssssssssssssssss+:`           --------------
      -+ssssssssssssssssssyyssss+-         OS: Ubuntu 24.04.1 LTS on Window
    .ossssssssssssssssssdMMMNysssso.       Kernel: 6.6.87.2-microsoft-stand
   /ssssssssssshdmmNNmmyNMMMMhssssss/      Uptime: 23 mins
  +ssssssssshmydMMMMMMMNddddyssssssss+     Packages: 603 (dpkg)
 /sssssssshNMMMyhhyyyyhmNMMMNhssssssss/    Shell: bash 5.2.21
.ssssssssdMMMNhsssssssssshNMMMdssssssss.   Theme: Adwaita [GTK3]
+ssssshhhyNMMNyssssssssssssyNMMMysssssss+  Icons: Adwaita [GTK3]
ossyNMMMNyMMhssssssssssssssshmmmhssssssso  Terminal: Relay(279)
ossyNMMMNyMMhssssssssssssssshmmmhssssssso  CPU: 12th Gen Intel i7-12700H (2
+ssssshhhyNMMNyssssssssssssyNMMMysssssss+  GPU: 6c4b:00:00.0 Microsoft Corp
.ssssssssdMMMNhsssssssssshNMMMdssssssss.   Memory: 409MiB / 7783MiB
 /sssssssshNMMMyhhyyyyhdNMMMNhssssssss/
  +ssssssssdmydMMMMMMMNddddyssssssss+
   /ssssssssssshdmNNNNmyNMMMMhssssss/
    .ossssssssssssssssssdMMMNysssso.
      -+sssssssssssssssssyyyssss+-
        `:+ssssssssssssssssss+:`
            .-/+oossssoo+/-.

tim@DellG15Tim:~/Documents$ cd ..
tim@DellG15Tim:~$ mkdir hello
tim@DellG15Tim:~$ cd hello/
tim@DellG15Tim:~/hello$ nano hello.sh
tim@DellG15Tim:~/hello$ chmod +x hello.sh
tim@DellG15Tim:~/hello$ ./hello.sh
Hello Tim Nijkamp, 591491!
tim@DellG15Tim:~/hello$ chmod 740 hello.sh
tim@DellG15Tim:~/hello$ ls -la
total 12
drwxr-xr-x 2 tim tim 4096 Jan 13 00:22 .
drwxr-x--- 8 tim tim 4096 Jan 13 00:21 ..
-rwxr----- 1 tim tim   44 Jan 13 00:22 hello.sh
tim@DellG15Tim:~/hello$
```

Assignment 5.6: View the contents of files

Relevant screenshots + motivation

Cat laat de inhoud van een bestand zien

Wc telt alle regels, woorden en bytes per bestand

Less laat een interactief tekst lezer zien.

Tail laat de laatste regels van een tekstbestand zien

Head laat het bovenste van een tekstbestand zien

Grep kijkt of er een regel voorkomt met de gegeven input

Het bestand heeft 12306 regels 107562 woorden en 607504 bytes

regel 490 en 1124 hebben het woord kingdom erin

```
tim@DellG15Tim: ~

Scandinavia. You may know the strict principles of her family. She is
herself the very soul of delicacy. A shadow of a doubt as to my conduct
would bring the matter to an end."

"And Irene Adler?"

"Threatens to send them the photograph. And she will do it. I know that
she will do it. You do not know her, but she has a soul of steel. She
has the face of the most beautiful of women, and the mind of the most
resolute of men. Rather than I should marry another woman, there are no
lengths to which she would not go—none."

"You are sure that she has not sent it yet?"

"I am sure."

"And why?"

"Because she has said that she would send it on the day when the
betrothal was publicly proclaimed. That will be next Monday."

"Oh, then we have three days yet," said Holmes with a yawn. "That is
very fortunate, as I have one or two matters of importance to look into
just at present. Your Majesty will, of course, stay in London for the
present?"

"Certainly. You will find me at the Langham under the name of the Count
Von Kramm."

"Then I shall drop you a line to let you know how we progress."

"Pray do so. I shall be all anxiety."

"Then, as to money?"

"You have _carte blanche_."

"Absolutely?"

"I tell you that I would give one of the provinces of my kingdom to
tim@DellG15Tim:~$
```

```
tim@DellG15Tim: ~
"Then, as to money?"

"You have _carte blanche_."

"Absolutely?"

"I tell you that I would give one of the provinces of my kingdom to
tim@DellG15Tim:~$ tail -n 1124 SherlockHolmes.txt
    "'JEPHRO RUCASTLE.'


"That is the letter which I have just received, Mr. Holmes, and my mind
is made up that I will accept it. I thought, however, that before
taking the final step I should like to submit the whole matter to your
consideration."

"Well, Miss Hunter, if your mind is made up, that settles the
question," said Holmes, smiling.

"But you would not advise me to refuse?"

"I confess that it is not the situation which I should like to see a
sister of mine apply for."

"What is the meaning of it all, Mr. Holmes?"

"Ah, I have no data. I cannot tell. Perhaps you have yourself formed
some opinion?"

"Well, there seems to me to be only one possible solution. Mr. Rucastle
seemed to be a very kind, good-natured man. Is it not possible that his
wife is a lunatic, that he desires to keep the matter quiet for fear
she should be taken to an asylum, and that he humours her fancies in
every way in order to prevent an outbreak?"

"That is a possible solution—in fact, as matters stand, it is the most
probable one. But in any case it does not seem to be a nice household
for a young lady."

"But the money, Mr. Holmes, the money!"
```

Assignment 5.7: Digital forensics

Relevant screenshots + motivation

Motorola telefoon

```
tim@DellG15Tim: ~
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
tim@DellG15Tim:~$ exif oldcar.jpg
EXIF tags in 'oldcar.jpg' ('Motorola' byte order):
--------------------+--------------------------------------------------------
---
Tag                 |Value
--------------------+--------------------------------------------------------
---
Manufacturer        |motorola
Model               |moto g(6) play
X-Resolution        |72
Y-Resolution        |72
Resolution Unit     |Inch
Software            |aljeter-user 9 PPPS29.55-35-18-7 6a0d0 release-keys
Date and Time       |2020:11:07 15:08:57
YCbCr Positioning   |Centered
Compression         |JPEG compression
X-Resolution        |72
Y-Resolution        |72
Resolution Unit     |Inch
Exposure Time       |1/33 sec.
F-Number            |f/2.0
Exposure Program    |Normal program
ISO Speed Ratings   |64
Exif Version        |Exif Version 2.2
Date and Time (Origi|2020:11:07 15:08:57
Date and Time (Digit|2020:11:07 15:08:57
Components Configura|Y Cb Cr -
Shutter Speed       |5.05 EV (1/33 sec.)
Aperture            |2.00 EV (f/2.0)
Brightness          |-1.00 EV (1.71 cd/m^2)
Exposure Bias       |0.00 EV
Maximum Aperture Val|2.00 EV (f/2.0)
Metering Mode       |Center-weighted average
Flash               |Flash did not fire, auto mode
Focal Length        |3.5 mm
Maker Note          |1719 bytes undefined data
FlashPixVersion     |FlashPix Version 1.0
Color Space         |sRGB
Pixel X Dimension   |4160
Pixel Y Dimension   |3120
```

De foto is in groningen gemaakt

Ja, ubuntu ziet het nog als een jpg

```
tim@DellG15Tim:~$ mv oldcar.jpg oldcar
tim@DellG15Tim:~$ ls
Documents  ITFundamentals  SherlockHolmes.txt  hello  oldcar
tim@DellG15Tim:~$ file oldcar
oldcar: JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segm
ent length 16, Exif Standard: [TIFF image data, big-endian, direntries=10, m
anufacturer=motorola, model=moto g(6) play, xresolution=160, yresolution=168
, resolutionunit=2, software=aljeter-user 9 PPPS29.55-35-18-7 6a0d0 release-
keys, datetime=2020:11:07 15:08:57, GPS-Data], baseline, precision 8, 4160x3
120, components 3
tim@DellG15Tim:~$
```

Assignment 5.8: Steganography

Relevant screenshots + motivation

```
tim@DellG15Tim: ~

 -e, --encryption        select encryption parameters
   -e <a>[<m>]|<m>[<a>]  specify an encryption algorithm and/or mode
   -e none               do not encrypt data before embedding
 -z, --compress          compress data before embedding (default)
   -z <l>                 using level <l> (1 best speed...9 best compression
)
 -Z, --dontcompress      do not compress data before embedding
 -K, --nochecksum        do not embed crc32 checksum of embedded data
 -N, --dontembedname     do not embed the name of the original file
 -f, --force             overwrite existing files
 -q, --quiet             suppress information messages
 -v, --verbose           display detailed information

extracting options:
 -sf, --stegofile        select stego file
   -sf <filename>        extract data from <filename>
 -p, --passphrase        specify passphrase
   -p <passphrase>       use <passphrase> to extract data
 -xf, --extractfile      select file name for extracted data
   -xf <filename>        write the extracted data to <filename>
 -f, --force             overwrite existing files
 -q, --quiet             suppress information messages
 -v, --verbose           display detailed information

options for the info command:
 -p, --passphrase        specify passphrase
   -p <passphrase>       use <passphrase> to get info about embedded data

To embed emb.txt in cvr.jpg: steghide embed -cf cvr.jpg -ef emb.txt
To extract embedded data from stg.jpg: steghide extract -sf stg.jpg
tim@DellG15Tim:~$ steghide extract -sf apple2.jpg
Enter passphrase:
wrote extracted data to "message.txt".
tim@DellG15Tim:~$ ls
Documents       SherlockHolmes.txt  email-base64.txt  hello        oldcar
ITFundamentals  apple2.jpg          email.gif         message.txt
tim@DellG15Tim:~$ cat message.txt
Hello class.
You have almost completed Week 5.

tim@DellG15Tim:~$
```

Assignment 5.9: Capture disk images

Make relevant screenshots + motivation:

- Proof that the Debian 13 server stored a back-up image of the Ubuntu 24.04 Desktop VM.
- Proof that you can restore the back-up image into an empty VM.

Ready? Save this file and export it as a pdf file with the name: **week5.pdf**